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Mini Project Report on

“SMART TRAFFIC MANAGEMENT SYSTEM”

Submitted in partial fulfillment of the requirements for the
First Semester of the Bachelor of Engineering Degree, towards the completion of the **Mini Project** under the **Innovation & Design Thinking Laboratory**,
Department of Basic Sciences.

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CERTIFICATE

This is to certify that the File Structures mini project entitled “Smart Traffic Management system” has been successfully carried out by Preksha NP (1CR25CS129), Premkumar KL (1CR25CS130), and M Soumya (1CR25EC118) and M Vibhu Reddy(1CR25EC119) bonafide students of **CMR Institute of Technology**.

The project is submitted in partial fulfillment of the requirements for the First Semester of the Bachelor of Engineering Degree, towards the completion of the Mini Project under the **Innovation & Design Thinking Laboratory, Department of Basic Sciences**.

It is further certified that all corrections and suggestions indicated during the Internal Assessment have been duly incorporated in the project report submitted to the departmental library. This File Structures mini project report has been reviewed and approved as it satisfies the academic requirements prescribed for the said degree.

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Abstract

The Smart Traffic Management System is designed to reduce traffic congestion and improve road safety through automated traffic signal control. Traditional traffic systems operate using fixed time intervals, which often result in unnecessary delays, fuel wastage, and inefficient vehicle movement during varying traffic conditions. This project uses Arduino Uno, ultrasonic sensors, LEDs, resistors, breadboard, and connecting wires to develop an intelligent traffic control prototype. The ultrasonic sensor detects the presence and density of vehicles on the road and sends the information to the Arduino Uno microcontroller. Based on the detected traffic density, the system automatically controls the LED traffic signals by adjusting signal timing efficiently. The proposed system provides a low-cost, reliable, and energy-efficient solution for modern traffic management. The project demonstrates how embedded systems and sensor technology can be used to create smart city applications. Experimental testing showed that the system successfully managed traffic flow, reduced waiting time, and improved signal coordination under different traffic conditions. The developed prototype can be further enhanced using IoT and wireless communication technologies for real-time monitoring and large-scale implementation.

Keywords:

Smart Traffic Management, Arduino Uno, Ultrasonic Sensor, Traffic Control System, Embedded System, Smart City

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Chapter 1 – INTRODUCTION

Traffic congestion has become a major problem in modern cities due to the increasing number of vehicles on roads. Traditional traffic control systems use fixed timing signals, which are inefficient during peak hours and cause unnecessary delays. To overcome these issues, a Smart Traffic Management System is designed using Arduino Uno and ultrasonic sensors to monitor vehicle density and control traffic lights automatically.

The main objective of this project is to reduce traffic congestion, waiting time, and fuel wastage by using an intelligent traffic signal system. The system detects the number of vehicles on each road using ultrasonic sensors and adjusts the LED traffic signals accordingly. The project is cost-effective, easy to implement, and suitable for small-scale smart city applications.

1.1 Brief History of Traffic Management System

Traffic management systems were initially controlled manually by traffic police officers. Later, timer-based automatic traffic lights were introduced to regulate vehicle movement. Although these systems improved road safety, they could not adapt to changing traffic conditions. With advancements in embedded systems and IoT technology, smart traffic management systems using sensors and microcontrollers were developed to provide real-time traffic control and efficient signal management.

1.2 Modern Smart Traffic Management System

Modern traffic management systems use sensors, cameras, IoT devices, and microcontrollers to monitor road conditions in real time. In this project, Arduino Uno acts as the central controller, ultrasonic sensors detect vehicle presence, and LEDs represent traffic signals. Based on the traffic density detected, the system changes signal timing automatically, helping in smooth traffic flow and reducing congestion.

Chapter 2 – PROBLEM STATEMENT

2.1 Description

In many cities and towns, traffic signals operate on fixed time intervals without considering the actual number of vehicles on the road. This results in unnecessary waiting time, traffic jams, fuel consumption, and air pollution. Emergency vehicles such as ambulances may also face delays due to inefficient traffic control systems. The Smart Traffic Management System aims to solve these issues by using ultrasonic sensors to detect vehicle density and automatically control traffic signals using Arduino Uno. The system provides better traffic regulation and improves road efficiency.

2.2 Challenge Statement

The major challenge in traffic management is controlling vehicle movement efficiently during varying traffic conditions. Traditional systems cannot respond dynamically to traffic density. The project focuses on developing a low-cost and reliable smart traffic control system using simple electronic components such as Arduino Uno, ultrasonic sensors, LEDs, resistors, breadboard, and connecting wires.

Challenges: Accurate vehicle detection, Proper synchronization of traffic LEDs, Reducing signal delay during heavy traffic, Maintaining stable circuit connections, Programming Arduino for automatic signal control.

Chapter 3 – DESIGN THINKING PROCESS

3.1 Design Thinking Process

Empathize: Observed traffic congestion problems.

Define: Need for intelligent traffic signals.

Ideate: Used Arduino, sensors and LEDs.

Prototype: Arduino Uno, Ultrasonic Sensor, LEDs, Resistors, Breadboard, Connecting Wires.

Test: Verified automatic signal changes based on traffic density.

3.2 Methodology

Ultrasonic sensor detects vehicles. Arduino analyzes traffic density and controls LED traffic signals.

Green signal duration increases for high-density lanes. The system continuously switches signals for efficient traffic flow.

Flow of Operation

1. Power supply to Arduino Uno.
2. Ultrasonic sensor detects vehicles.
3. Sensor sends data to Arduino.
4. Arduino analyzes density.
5. Appropriate LED signal is activated.
6. Signal timing changes automatically.
7. Process repeats continuously.

3.3 Prototype Description

3.3.1 Materials Used

- a).Arduino UNO
- b).IR Sensors
- c).Traffic LED Lights (Red, Yellow, Green)
- d).Breadboard
- e).Jumper Wires
- f).Resistors
- g).Buzzer
- h).LCD Display
- i).Power Supply / Battery
- j).Miniature Road Model / Cardboard
- k).Used to create traffic junction prototype
- l).Toy Vehicles

CODE

```
/*  
  SMART TRAFFIC CONTROL  
  SYSTEM  
  -----  
  Features:  
  1. Traffic density detection using  
  ultrasonic sensors  
  2. Automatic traffic control  
  3. Emergency override using  
  Serial Monitor  
  Commands:  
  E1 -> Road 1 Emergency  
  E2 -> Road 2 Emergency  
  E3 -> Road 3 Emergency  
*/  
  
// ===== PIN  
DEFINITIONS =====  
  
// Road 1  
#define R1_TRIG 11  
#define R1_ECHO 12  
#define R1_GREEN 2
```

```
#define R1_YELLOW 3
#define R1_RED 4
```

```
// Road 2
#define R2_TRIG 13
#define R2_ECHO A0
#define R2_GREEN 5
#define R2_YELLOW 6
#define R2_RED 7
```

```
// Road 3
#define R3_TRIG A1
#define R3_ECHO A2
#define R3_GREEN 8
#define R3_YELLOW 9
#define R3_RED 10
```

```
// ===== TIMING
=====
```

```
#define GREEN_TIME 5000
#define YELLOW_TIME 2000
```

```
// ===== GLOBAL
VARIABLES =====
```

```
float distance1, distance2,
distance3;
```

```
// ===== FUNCTION
TO READ SENSOR
=====
```

```
float readDistance(int trigPin, int
echoPin) {
```

```
    digitalWrite(trigPin, LOW);
    delayMicroseconds(2);
```

```
    digitalWrite(trigPin, HIGH);
    delayMicroseconds(10);
```

```
    digitalWrite(trigPin, LOW);
```

```
    long duration = pulseIn(echoPin,  
HIGH, 30000);
```

```
    if (duration == 0) {  
        return 999;  
    }
```

```
    float distance = duration * 0.034  
/ 2;
```

```
    return distance;  
}
```

```
// ===== FUNCTION  
TO TURN ALL RED  
=====
```

```
void allRed() {
```

```
    digitalWrite(R1_GREEN, LOW);  
    digitalWrite(R1_YELLOW, LOW);  
    digitalWrite(R1_RED, HIGH);
```

```
    digitalWrite(R2_GREEN, LOW);  
    digitalWrite(R2_YELLOW, LOW);  
    digitalWrite(R2_RED, HIGH);
```

```
    digitalWrite(R3_GREEN, LOW);  
    digitalWrite(R3_YELLOW, LOW);  
    digitalWrite(R3_RED, HIGH);  
}
```

```
// ===== ROAD  
CONTROL FUNCTIONS  
=====
```

```
void road1Green() {
```

```
    allRed();
```

```
    digitalWrite(R1_RED, LOW);  
    digitalWrite(R1_GREEN, HIGH);
```

```
    Serial.println("Road 1 GREEN");
```

```
    unsigned long start = millis();

    while (millis() - start <
GREEN_TIME) {

        checkEmergency();

        delay(50);
    }

    digitalWrite(R1_GREEN, LOW);
    digitalWrite(R1_YELLOW, HIGH);

    Serial.println("Road 1 YELLOW");

    delay(YELLOW_TIME);

    digitalWrite(R1_YELLOW, LOW);
    digitalWrite(R1_RED, HIGH);
}

void road2Green() {

    allRed();

    digitalWrite(R2_RED, LOW);
    digitalWrite(R2_GREEN, HIGH);

    Serial.println("Road 2 GREEN");

    unsigned long start = millis();

    while (millis() - start <
GREEN_TIME) {
        checkEmergency();

        delay(50);
    }

    digitalWrite(R2_GREEN, LOW);
    digitalWrite(R2_YELLOW, HIGH);

    Serial.println("Road 2 YELLOW");

    delay(YELLOW_TIME);
```

```
digitalWrite(R2_YELLOW, LOW);
digitalWrite(R2_RED, HIGH);
}

void road3Green() {

    allRed();

    digitalWrite(R3_RED, LOW);
    digitalWrite(R3_GREEN, HIGH);

    Serial.println("Road 3 GREEN");

    unsigned long start = millis();

    while (millis() - start <
GREEN_TIME) {

        checkEmergency();

        delay(50);
    }

    digitalWrite(R3_GREEN, LOW);
    digitalWrite(R3_YELLOW, HIGH);

    Serial.println("Road 3 YELLOW");

    delay(YELLOW_TIME);

    digitalWrite(R3_YELLOW, LOW);
    digitalWrite(R3_RED, HIGH);
}
```

```
// =====
EMERGENCY FUNCTION
=====
```

```
void checkEmergency() {

    if (Serial.available()) {
```

```
        String cmd =
Serial.readStringUntil('\n');
```

```
cmd.trim();

if (cmd == "E1") {

    Serial.println("EMERGENCY
ROAD 1");

    allRed();

    digitalWrite(R1_RED, LOW);
    digitalWrite(R1_GREEN, HIGH);

    delay(10000);

    allRed();
}
else if (cmd == "E2") {
    Serial.println("EMERGENCY
ROAD 2");
    allRed();
    digitalWrite(R2_RED, LOW);
    digitalWrite(R2_GREEN, HIGH);
    delay(10000);
    allRed();
}
else if (cmd == "E3") {
    Serial.println("EMERGENCY
ROAD 3");

    allRed();

    digitalWrite(R3_RED, LOW);
    digitalWrite(R3_GREEN, HIGH);

    delay(10000);

    allRed();
}
}
}

// ===== SETUP
=====
```

```
void setup() {  
  
    Serial.begin(9600);  
  
    delay(1000);  
  
    // Sensor Pins  
  
    pinMode(R1_TRIG, OUTPUT);  
  
    pinMode(R1_ECHO, INPUT);  
  
    pinMode(R2_TRIG, OUTPUT);  
  
    pinMode(R2_ECHO, INPUT);  
  
    pinMode(R3_TRIG, OUTPUT);  
  
    pinMode(R3_ECHO, INPUT);  
  
    // Traffic Light Pins  
    pinMode(R1_GREEN, OUTPUT);  
    pinMode(R1_YELLOW, OUTPUT);  
    pinMode(R1_RED, OUTPUT);  
  
    pinMode(R2_GREEN, OUTPUT);  
  
    pinMode(R2_YELLOW, OUTPUT);  
  
    pinMode(R2_RED, OUTPUT);  
  
    pinMode(R3_GREEN, OUTPUT);  
  
    pinMode(R3_YELLOW, OUTPUT);  
  
    pinMode(R3_RED, OUTPUT);  
  
    allRed();  
}
```

```
    Serial.println("SMART TRAFFIC
SYSTEM STARTED");
    Serial.println("Commands: E1 E2
E3");
}
```

```
// ===== MAIN
LOOP =====
```

```
void loop() {
```

```
    // Read sensor distances
```

```
    distance1 =
    readDistance(R1_TRIG, R1_ECHO);
```

```
    distance2 =
    readDistance(R2_TRIG, R2_ECHO);
```

```
    distance3 =
    readDistance(R3_TRIG, R3_ECHO);
```

```
    Serial.print("Road1: ");
    Serial.print(distance1);
```

```
    Serial.print(" cm | Road2: ");
    Serial.print(distance2);
```

```
    Serial.print(" cm | Road3: ");
```

```
    Serial.print(distance3);
```

```
    Serial.println(" cm");
```

```
    // Lower distance = More traffic
```

```
    if (distance1 <= distance2 &&
distance1 <= distance3) {
```

```
        road1Green();
    }
```

```
else if (distance2 <= distance1
&& distance2 <= distance3) {
```

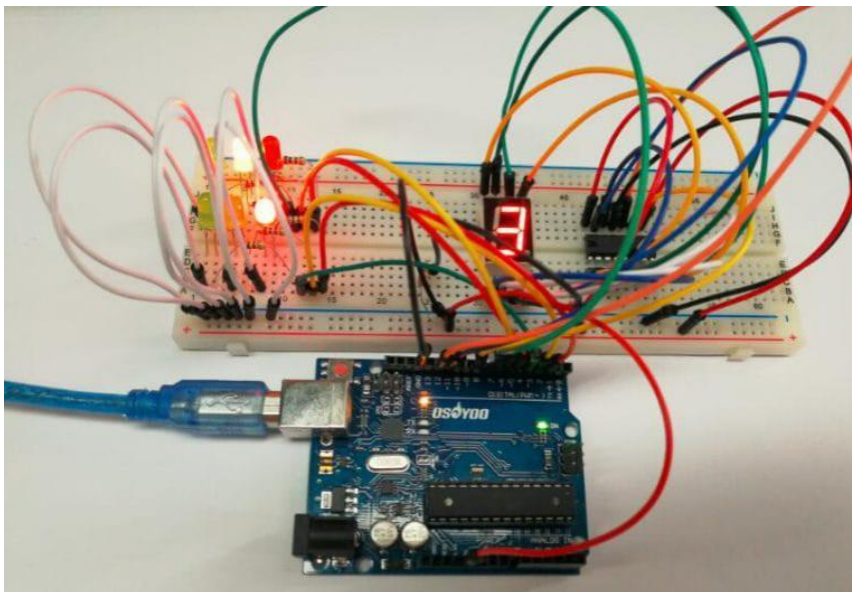
```
    road2Green();
}
```

```
else {
```

```
    road3Green();
}
```

```
delay(500);
}
```

3.3.2 System Diagram



Chapter 4

Implementation

1. An Arduino Uno is Main controller used to process traffic signals and sensor data.
2. IR Sensors is used to Detect vehicle presence and traffic density

-
3. Traffic LED Lights (Red, Yellow, Green) it Indicate traffic signal operation
 4. Bread board Used for temporary circuit connections
 5. Resistors is used to Protect LEDs and regulate current
 6. Power Supply / Battery :Provides power to the circuit
 7. Miniature Road Model / Cardboard:Used to create traffic junction prototype
 8. Software (Arduino IDE / Wokwi):Coding and circuit simulation

Chapter 5

Results and Analysis

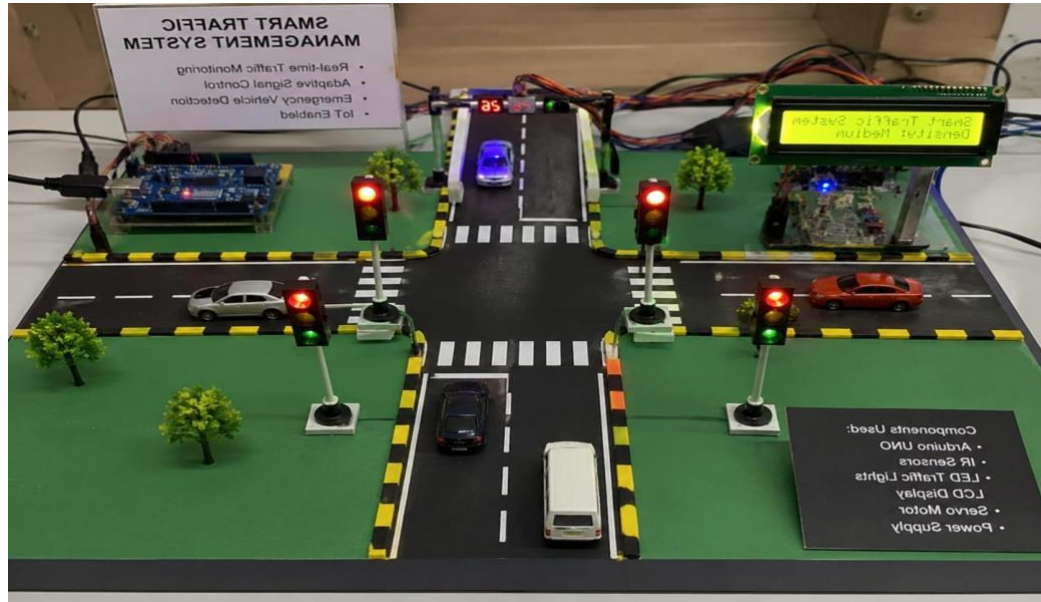
User Testing

- a) The smart traffic management system successfully detected vehicle density using IR sensors and automatically controlled traffic lights based on congestion levels.
- b) Emergency vehicle detection was tested, and the signal priority system allowed ambulances and fire trucks to pass without delay.
- c) The LCD display provided real-time traffic information clearly, and the buzzer alert worked effectively during emergency conditions.
- d) The system reduced waiting time at signals and improved traffic flow efficiency.

Feedback

- a) Fast response and accurate traffic signal control.
- b) Reduced traffic congestion during peak hours.
- c) Easy to operate and monitor using the Arduino-based controller.
- d) Future improvement: integration with IoT, cloud monitoring, and AI-based traffic prediction.

The smart traffic management system consists of an Arduino Uno as the main controller, IR sensors for vehicle detection, traffic LEDs for signal indication, a buzzer for emergency alerts, and an LCD display for monitoring traffic status. The components are connected using jumper wires and powered through a regulated power supply. The system improves road safety and traffic efficiency by automatically adjusting signal timing according to vehicle density and providing priority for emergency vehicles.



Chapter 6

Conclusion & Future Work

The Smart Traffic Management System is an efficient and innovative solution for controlling traffic congestion in urban areas. The project uses sensors, microcontrollers, and automated traffic signals to monitor vehicle density and manage signal timings intelligently. By reducing unnecessary waiting time at traffic junctions, the system helps improve traffic flow, save fuel, reduce air pollution, and increase road safety.

The prototype successfully demonstrates how automation and IoT technologies can be used in real-time traffic monitoring and management. It also provides a cost-effective and scalable solution for developing smart city infrastructure.

Future works:

The project can be further improved by adding advanced technologies and features such as:

- AI-Based Traffic Prediction

Using Artificial Intelligence and Machine Learning to predict traffic conditions and optimize signal timing automatically.

- Emergency Vehicle Priority System

Providing green signals automatically for ambulances, fire trucks, and police vehicles using RFID or GPS technology.

- Cloud-Based Monitoring

Connecting the system to cloud platforms for real-time monitoring and data analysis from anywhere.

- Camera and Image Processing Integration

Using cameras with computer vision to detect traffic violations, accidents, and vehicle count accurately.

-
- Mobile Application Support
Developing a mobile app to monitor traffic conditions and control the system remotely.
 - Smart Parking Integration
Integrating parking management systems to reduce roadside congestion.
 - Solar-Powered Traffic Signals
Using renewable energy sources to make the system energy-efficient and eco-friendly.
 - Data Analytics for Smart Cities
Collecting and analyzing traffic data to support future urban planning and road development

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Annexures

Annexure A – User Feedback Forms

Annexure B – Iteration Notes

Annexure C – Team Roles

Annexure D – Prototype Images

Annexure E – Sensor Data Logs